

Lecture 3 Review Handout

Statistics 367-1-4361

Lecture Overview

Lecture 3 moves from analytic posterior calculation to **computational Bayesian inference**. It introduces graphical models and probabilistic programming languages, then explains why we sample from posteriors and how PyMC stores those draws. The lecture next uses posterior draws for visualization, point estimation, and HDIs. It then shifts to a frequentist lens, covering repeated-sampling ideas, estimators, maximum likelihood, and confidence intervals as an alternative way to quantify uncertainty.

Lecture Sections

3.1 — Graphical models

This section introduces graphical models as compact representations of how a probabilistic model is structured. The main goal is not aesthetics but clarity: a graph should show what counts as data, what counts as a distribution, and how parameters connect to observations. The slides also emphasize that there is no single correct notation, only clearer or less clear ones.

New Definitions and Terms

- **Graphical model** — a diagram showing the structure of a probabilistic model
- **Node** — a graphical element representing data, a distribution, or an equation
- **Arrow** — a graphical indication that one quantity helps determine another
- **DAG** — a directed acyclic graph, one common style for representing probabilistic models
- **Plate notation** — a graphical convention for repeated structure

Figure

Representing models using graphs

- We will represent our models with diagrams
 - Data at the bottom
 - z heads
 - Likelihood in the middle
 - This is our model of how the data was produced
 - N tosses of a coin with probability θ of getting heads
 - Prior at the top
 - This reflects our uncertainty about θ
 - When the experiment began

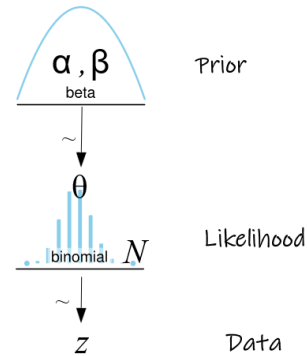


Figure 1: Slide 4 — Representing models using graphs

3.2 — Probabilistic programming languages

This section explains why Bayesian inference became computational rather than purely analytic. As models became more realistic, exact posterior calculation often became impossible, so probabilistic programming languages were developed to let us specify models and then compute with them. PyMC is introduced as the course PPL, and the lecture shows that the same model can be written as equations, a graph, or code.

New Definitions and Terms

- **Probabilistic programming language (PPL)** — a programming system for specifying probabilistic models and carrying out inference
- **PyMC** — the Python PPL used in the course
- **Graphical visualizer** — a tool for displaying the dependency structure of a PyMC model

Figure

The PyMC graphical visualizer

- Visualize any PyMC model
 - Shows: dimensionality, distributions, and dependencies

```
pm.model_to_graphviz(our_first_model)
```

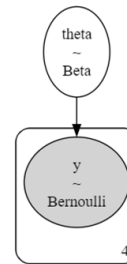


Figure 2: Slide 11 — The PyMC graphical visualizer

3.3 — Sampling

This section gives the main computational motivation for the lecture: many posteriors are too hard to calculate analytically and too large to approximate well by a grid. Sampling solves this by generating draws from the posterior distribution, with more draws giving a more stable approximation. The section also distinguishes **draws** from **chains**, and warns that the word *sample* is overloaded in statistics.

New Definitions and Terms

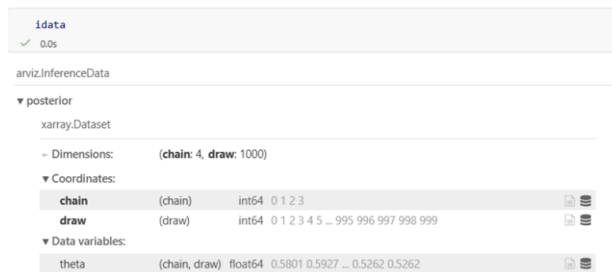
- **Draw** — one set of parameter values generated from the posterior distribution
- **Chain** — a sequence of posterior draws generated by one run of the sampler
- **Posterior sampling** — generating draws that approximate the posterior distribution
- **InferenceData object** — the object PyMC returns to store posterior draws and related outputs

Figure

posterior has the samples

- One variable: θ
- Two dimensions: chain and draw
- 4,000 draws

A draw is one set of parameters sampled from the posterior distribution
A chain is a series of draws



idata	
✓	0.0s
arviz.InferenceData	
▼ posterior	
xarray.Dataset	
- Dimensions: (chain: 4, draw: 1000)	
▼ Coordinates:	
chain	(chain) int64 0 1 2 3
draw	(draw) int64 0 1 2 3 4 5 ... 995 996 997 998 999
▼ Data variables:	
theta	(chain, draw) float64 0.5801 0.5927 ... 0.5262 0.5262

Figure 3: Slide 19 — posterior has the samples

3.4 — Plotting the posterior distribution

This section introduces ArviZ as the main plotting and summary tool for posterior draws. The emphasis is practical: once we have posterior draws, we need ways to inspect them visually and summarize them numerically. The trace plot and posterior plot are presented as the first standard views of sampled Bayesian results.

New Definitions and Terms

- **ArviZ** — the package used for analyzing and plotting Bayesian posterior results
- **Trace plot** — a plot of sampled draws across chains and iterations
- **Posterior plot** — a plot summarizing the posterior distribution of a parameter

Figure

Arviz summary of the posterior

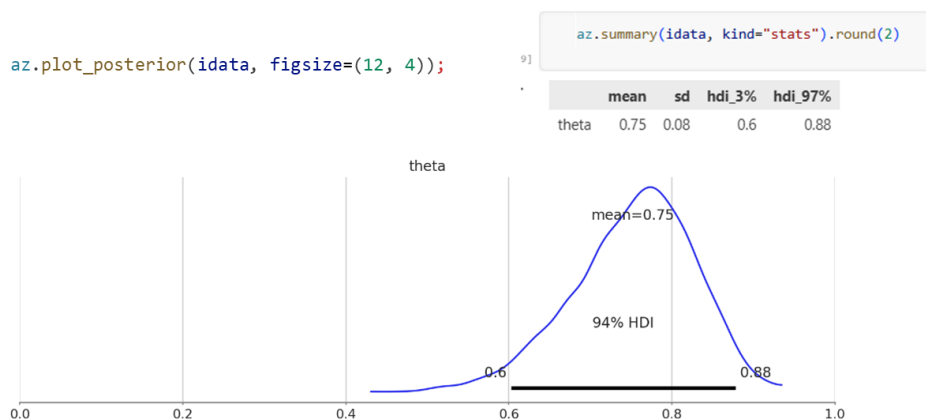


Figure 4: Slide 24 — Arviz summary of the posterior

3.5 — Estimation

This section asks how to turn a posterior distribution into a single best guess. Using simulated data, the lecture compares different estimators and shows that they can differ in both **bias** and **variance**. The main message is that point estimation is a summary of the posterior, not a replacement for the full posterior.

New Definitions and Terms

- **Estimator** — a rule for producing a numerical guess from data or from a posterior
- **Bias** — systematic deviation of an estimator from the true value
- **Variance** — the amount an estimator changes across repeated datasets
- **Bias-variance tradeoff** — the fact that lower bias can come with higher variance, or vice versa
- **Posterior mean** — the average of the posterior distribution
- **Posterior mode** — the value with the highest posterior density

Figure

The distribution of the estimators

- Across multiple simulated datasets
 - With the same true value of the parameter
 - The distributions are affected by having discrete data
 - We can see differences more clearly by zooming in

Estimator	Bias	Std dev
Posterior mean	-0.0195	0.0676
Posterior median	-0.0136	0.0691
Posterior mode	-0.0012	0.0739

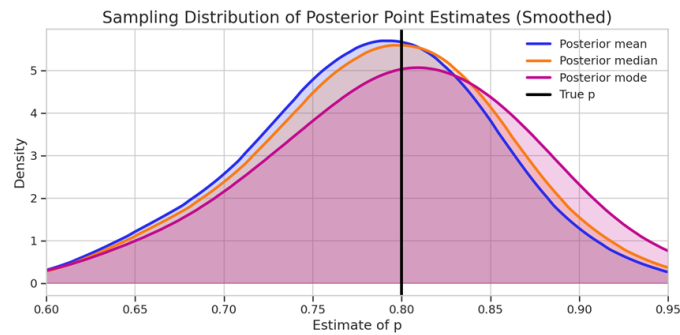


Figure 5: Slide 29 — The distribution of the estimators

3.6 — HDI

This section introduces the **highest density interval** as a compact summary of posterior uncertainty. An HDI contains a fixed amount of posterior probability while prioritizing the most credible values, so in the unimodal univariate case it is the shortest interval with that probability mass. The lecture also stresses that HDI probability and repeated-sampling coverage are not the same thing.

New Definitions and Terms

- **HDI** — the highest density interval; an interval containing the most credible posterior values
- **Coverage** — the fraction of repeated datasets for which an interval would contain the true parameter
- **Disconnected HDI** — an HDI that splits into separate regions when the posterior is multi-peaked

Figure

The logic of the HDI

- Unimodal
- Univariate
- HDI is the shortest interval

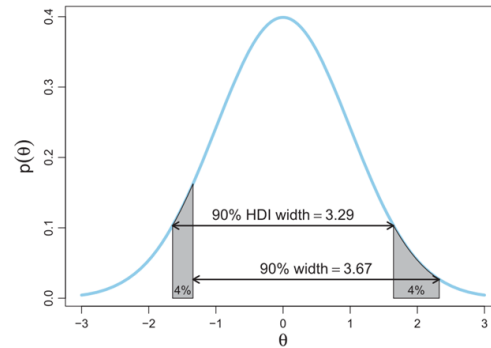


Figure 6: Slide 36 — The logic of the HDI

3.7 — Frequentist vs Bayesian philosophies

This section presents the conceptual split between Bayesian and frequentist reasoning. The Bayesian view treats data as updating our uncertainty about parameters, while the frequentist view treats uncertainty as arising from repeated random datasets generated by a procedure. The key point is that these approaches are answering different questions, not merely using different notation.

New Definitions and Terms

- **Bayesian uncertainty** — uncertainty about parameters or models given the data and assumptions
- **Frequentist uncertainty** — uncertainty defined through repeated sampling behavior of a procedure
- **Procedure-based inference** — inference judged by how a method behaves across repeated uses

Figure

Bayesians and frequentists ask different questions

- Bayesian
 - Given the model and the data, what should we believe about the parameter?
- Frequentist
 - If we repeat the procedure many times, how will the estimator behave?
- This course is “Bayesian first”
 - Learn about frequentist methods
 - But understand them from a Bayesian perspective

	Bayesian	Frequentist
Object of uncertainty	Parameters	Repeated sample behavior
Probability	Aleatory + epistemic	Aleatory
Goal of analysis	Posterior distribution	Estimator
Assessing uncertainty	Posterior distribution	Sampling interval

Figure 7: Slide 42 — Bayesians and frequentists ask different questions

3.8 — Frequentist estimation

This section develops the frequentist idea of an estimator as a statistic whose quality is judged across repeated experiments. The lecture compares several estimators of a mean and shows that their behavior depends strongly on data quality, sample size, and outliers. The sampling distribution is the central object here because it tells us how an estimator behaves under replication.

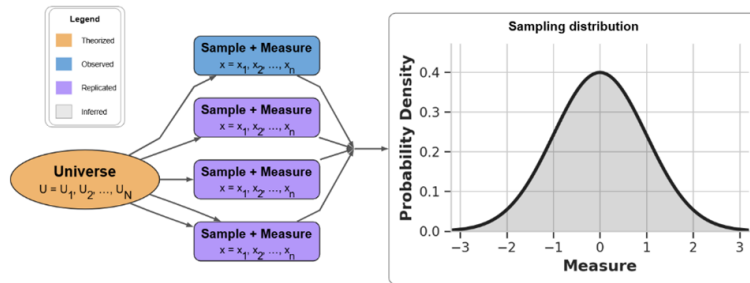
New Definitions and Terms

- **Statistic** — any quantity calculated from data
- **Estimator** — a statistic used to guess a real parameter value
- **Sampling distribution** — the distribution of an estimator across repeated datasets
- **Reduced mean** — a mean computed after trimming extreme observations

Figure

Sampling distribution of the statistic

- The distribution we expect for the estimator under replications



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Figure 8: Slide 47 — Sampling distribution of the statistic

Sampling distribution for estimator uncertainty

- Measure uncertainty of mean, median and mode

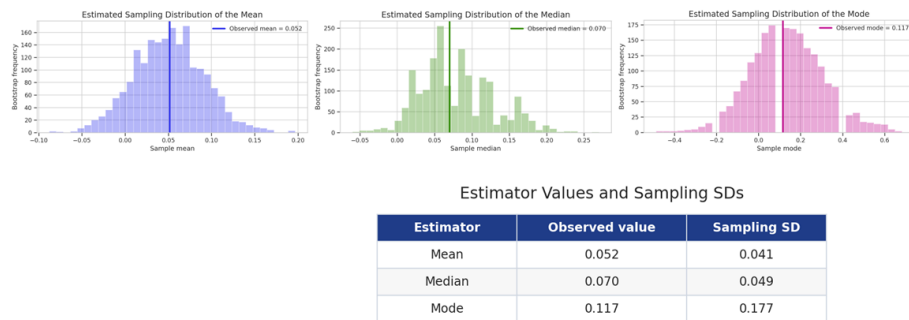


Figure 9: Slide 50 — Sampling distribution for estimator uncertainty

3.9 — Maximum likelihood

This section introduces the **maximum likelihood estimate** as a model-based frequentist estimator. The MLE chooses the parameter value that makes the observed data most plausible under the likelihood. The lecture then interprets MLE from a Bayesian perspective: it resembles taking a flat prior, using the likelihood as the posterior shape, and then collapsing that distribution to a single point.

New Definitions and Terms

- **Maximum likelihood estimate (MLE)** — the parameter value that maximizes the likelihood
- **Consistency** — the property that an estimator approaches the true value as sample size grows
- **Efficiency** — low variance relative to other estimators, especially in large samples
- **Model misspecification** — using a model that does not match the real data-generating process

Figure

What is the MLE?

Likelihood

- MLE is the maximum of the likelihood
 - As a function of the parameter
 - Requires a model
- Why MLE?
 - The parameter that makes our data easiest to believe

$$p(\text{data} \mid \text{parameters})$$

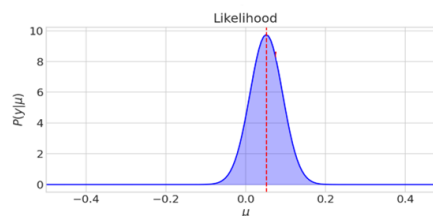


Figure 10: Slide 52 — What is the MLE?

3.10 — Confidence intervals

This section defines the confidence interval through the sampling distribution, not through posterior probability. The lecture builds lower and upper tail statistics, then defines a confidence interval as the set of parameter values for which the observed estimate is not too unusual. The slides end by contrasting CI logic with HDI logic and by warning against the very common mistake of treating a confidence interval as a probability statement about the parameter.

New Definitions and Terms

- **Confidence interval (CI)** — an interval-valued procedure with a stated repeated-sampling coverage rate
- **Tail statistic** — a boundary defined by a tail probability in the sampling distribution
- **Coverage probability** — the long-run proportion of intervals that contain the true parameter

Formulas

$$[p(L(\hat{y}) \leq H(\hat{y})) = 0.96]$$

A 96% confidence interval is defined by lower and upper bounds whose repeated-sampling procedure contains the true parameter 96% of the time.

Figure

Definition of the confidence interval

- If we can define two functions: $L(y^*)$ and $H(y^*)$
 - Of any dataset y^*
 - Such that $p(L(y^*) \leq \theta \leq H(y^*)) = 0.96$
 - The probability is over different datasets
 - Equality holds no matter what the true θ
- Then:
 - Given a real dataset y
 - The interval $[L(y), H(y)]$ is the 96% confidence interval

Figure 11: Slide 60 — Definition of the confidence interval

3.11 — Next time

The lecture closes by pointing out that drawing posterior samples is itself a hard problem. If there is no direct random-number generator for the posterior, we need an algorithm that moves through parameter space in a way that matches posterior probability. That motivates the next lecture on MCMC.

New Definitions and Terms

- **MCMC** — a family of algorithms for generating draws from difficult posterior distributions
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